

Applicability of the Ditlevsen and Samson Gaussian approximation to Dacca

I don't think the Gaussian transition approximation used by Ditlevsen and Samson (DS) carries over directly to Dacca. Hypoellipticity itself is not the issue. It is well known that a process can have a rank-deficient instantaneous diffusion and a smooth positive-time transition density. Here, the problem is the number of stochastic directions recovered by the order-1.5 approximation.

In DS equations (2) through (4), the state is $(V, U^\top)^\top \in \mathbb{R}^{p+1}$, where V is scalar and $U \in \mathbb{R}^p$. The U coordinates are driven by p independent Brownian motions through a positive diagonal matrix Γ . The diffusion rank is therefore $p = d - 1$, leaving one direction to be supplied through the drift. Write $G = \Gamma\Gamma^\top$ and $r = \partial_u a$. The leading covariance in their equation (33) is

$$\Sigma_\Delta = \begin{bmatrix} \Delta^3 r G r^\top / 3 & \Delta^2 r G / 2 \\ \Delta^2 G r^\top / 2 & \Delta G \end{bmatrix}.$$

The Schur complement of ΔG is

$$\frac{\Delta^3}{3} r G r^\top - \frac{\Delta^2}{2} r G (\Delta G)^{-1} \frac{\Delta^2}{2} G r^\top = \frac{\Delta^3}{12} r G r^\top > 0$$

under condition (C1). Hence

$$\det \Sigma_\Delta = \frac{\Delta^{p+3}}{12} \det(G) r G r^\top > 0.$$

Equation (33) is stated to leading order, so this calculation gives the small-step nondegeneracy mechanism. In the harmonic-oscillator example, the displayed covariance has determinant $\sigma^4 \Delta^4 / 12$ for every $\Delta > 0$.

Dacca instead has the active state $X = (S, I, R^1, R^2, R^3, M)^\top \in \mathbb{R}^6$ and one Brownian vector,

$$g(x) = \sigma S I e, \quad e = (-1, 1, 0, 0, 0, 0)^\top.$$

Its diffusion rank is one when $SI \neq 0$ and zero on the boundary. Propagating this scalar-noise direction through the first drift derivative gives

$$Q_h^{(1.5)} = A_2 H_2 A_2^\top, \quad A_2 = [h^{1/2} g, h^{3/2} D b g], \quad H_2 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}.$$

It follows that $\text{rank } Q_h^{(1.5)} \leq 2 < 6$ for every $h > 0$. The state-dependent diffusion coefficient does not supply more directions because $g = qe$. That is, the Dg and Milstein terms remain parallel to e , while $D b g$ adds only the first bracket direction. So $Q_h^{(1.5)}$ has zero determinant, and so the inverse, log determinant, and six-dimensional Gaussian density are undefined. We would have to add a ridge correction. At the manuscript parameter and state point, the matrix formed from $g, [b^S, g], \dots, \text{ad}_{b^S}^5 g$, where b^S is the Stratonovich drift, first reaches rank six when the fifth nested drift bracket is included. Here “depth” counts bracket nesting, not the number of numerical substeps. This suggests that the true positive-time transition may have a smooth density, but it does not make the first-bracket Gaussian nonsingular.

S. Ditlevsen and A. Samson, *Hypoelliptic diffusions: filtering and inference from complete and partial observations*, equations (2) through (4), (33), (38), condition (C1), and Section 5.1. arXiv:1707.04235.